

Accelerator ML Living Review

Summary Statistics

per_year: 12
per_category: 15
per_venue/journal: 37
per_keyword: 15
monthly_trends: 72

Papers

A Supervised Machine Learning Framework for Multipactor Breakdown Prediction in High-Power Radio Frequency Devices and Accelerator Components: A Case Study in Planar Geometry

Asif Iqbal, John Verboncoeur, Peng Zhang (2025)
arXiv

Geoff: The Generic Optimization Framework & Frontend for Particle Accelerator Controls

Madysa, Penelope, Appel, Sabrina, Kain, Verena, Schenk, Michael (2025)
SoftwareX

Acceleration of Multi-Scale LTS Magnet Simulations with Neural Network Surrogate Models

Louis Denis, Julien Dular, Vincent Nuttens, Mariusz Wozniak, Benoît Vanderheyden, Christophe Geuzaine (2025)
arXiv

Batch spacing optimization by reinforcement learning

Remta, Matthias, Velotti, Francesco, Rezagholi, Sharwin (2025)
Phys.Rev.Accel.Beams

Optimisation of the Accelerator Control by Reinforcement Learning: A Simulation-Based Approach

Anwar Ibrahim, Denis Derkach, Alexey Petrenko, Fedor Ratnikov, Maxim Kaledin (2025)
arXiv

Optimizing Beam-Plasma Interactions Through Jitter Analysis Using Start-to-End Simulations

Hwang, Robin (2024)
arXiv

Virtual Pulse Reconstruction Diagnostic for Single-Shot Measurement of Free Electron Laser Radiation Power

Till Korten, Vladimir Rybnikov, Peter Steinbach, Najmeh Mirian (2024)
Phys.Rev.Accel.Beams

Data-driven gradient optimization for field emission management in a superconducting radio-frequency linac

Goldenberg, Steven, Ahammed, Kawser, Carpenter, Adam, Li, Jiang, Suleiman, Riad, Tennant, Chris (2024)
Phys.Rev.Accel.Beams

Data-Driven Discovery of Beam Centroid Dynamics

Pocher, Liam A., Haber, Irving, Antonsen, Thomas M., O'Shea, Patrick G. (2024)
arXiv

Design and development of advanced Al-Ti-V alloys for beampipe applications in particle accelerators

Kamaljeet Singh, Kangkan Goswami, Raghunath Sahoo, Sumanta Samal (2025)
arXiv

Machine Learning for Reducing Noise in RF Control Signals at Industrial Accelerators
M. Henderson, J. P. Edelen, J. Einstein-Curtis, C. C. Hall, J. A. Diaz Cruz, A. L. Edelen (2024)
JINST

Beamline Steering Using Deep Learning Models
Allen, Dexter, Kante, Isaac, Bohler, Dorian (2024)
arXiv

Beam-based Identification of Magnetic Field Errors in a Synchrotron using Deep Lie Map Networks
Conrad Caliri, Adrian Oeftiger, Oliver Boine-Frankenheim (2024)
Phys.Rev.Accel.Beams

Linac_Gen: integrating machine learning and particle-in-cell methods for enhanced beam dynamics at Fermilab
Abhishek Pathak (2024)
arXiv

Automated Anomaly Detection on European XFEL Klystrons
Antonin Sulc, Annika Eichler, Tim Wilksen (2024)
arXiv

Accelerator beam phase space tomography using machine learning to account for variations in beamline components
Andrzej Wolski, Diego Botelho, David Dunning, Amelia E. Pollard (2024)
arXiv

Accelerating Cavity Fault Prediction Using Deep Learning at Jefferson Laboratory
Monibor Rahman, Adam Carpenter, Khan Iftekharuddin, Chris Tennant (2024)
arXiv

Machine-learning approach for operating electron beam at KEK S_e^-/e^+ injector Linac
Gaku Mitsuka, Shinnosuke Kato, Naoko Iida, Takuya Natsui, Masanori Satoh (2024)
arXiv

Cheetah: Bridging the Gap Between Machine Learning and Particle Accelerator Physics with High-Speed, Differentiable Simulations
Jan Kaiser, Chenran Xu, Annika Eichler, Andrea Santamaria Garcia (2024)
arXiv

Robust Errant Beam Prognostics with Conditional Modeling for Particle Accelerators
Kishansingh Rajput, Malachi Schram, Willem Blokland, Yasir Alanazi, Pradeep Ramuhalli, Alexander Zhukov, Charles Peters, Ricardo Vilalta (2024)
arXiv

Machine Learning For Beamline Steering
Isaac Kante (2023)
arXiv

Variational Autoencoders for Noise Reduction in Industrial LLRF Systems
J. P. Edelen, M. J. Henderson, J. Einstein-Curtis, C. C. Hall, J. A. Diaz Cruz, A. L. Edelen (2023)
arXiv

Resilient VAE: Unsupervised Anomaly Detection at the SLAC Linac Coherent Light Source
Ryan Humble, William Colocho, Finn O'Shea, Daniel Ratner, Eric Darve (2023)
arXiv

Time-drift Aware RF Optimization with Machine Learning Techniques
R. Sharankova, M. Mwaniki, K. Seiya, M. Wesley (2023)
arXiv

Forecasting Particle Accelerator Interruptions Using Logistic LASSO Regression

Sichen Li, Jochem Snuverink, Fernando Perez-Cruz, Andreas Adelmann (2023)
arXiv

Learning Electron Bunch Distribution along a FEL Beamline by Normalising Flows

Anna Willmann, Jurjen Couperus Cabada, Yen-Yu Chang, Richard Pausch, Amin Ghaith, Alexander Debus, Arie Irman, Michael Bussmann, Ulrich Schramm, Nico Hoffmann (2023)
arXiv

Physics-constrained 3D Convolutional Neural Networks for Electrodynamics

Alexander Scheinker, Reemu Pokharel (2023)
arXiv

Identification of Magnetic Field Errors in Synchrotrons based on Deep Lie Map Networks

Conrad Caliri, Adrian Oeftiger, Oliver Boine-Frankenheim (2023)
arXiv

Applications of Differentiable Physics Simulations in Particle Accelerator Modeling

Ryan Roussel, Auralee Edelen (2022)
arXiv

Prior-mean-assisted Bayesian optimization application on FRIB Front-End tuning

Kilean Hwang, Tomofumi Maruta, Alexander Plastun, Kei Fukushima, Tong Zhang, Qiang Zhao, Peter Ostroumov, Yue Hao (2022)
arXiv

Fault Prognosis in Particle Accelerator Power Electronics Using Ensemble Learning

Majdi I. Radaideh, Chris Pappas, Mark Wezensky, Pradeep Ramuhalli, Sarah Cousineau (2022)
arXiv

Machine learning-based analysis of experimental electron beams and gamma energy distributions

M. Yadav, M. Oruganti, S. Zhang, B. Naranjo, G. Andonian, Y. Zhuang, Ö. Apsimon, C. P. Welsch, J. B. Rosenzweig (2023)
arXiv

Review of Time Series Forecasting Methods and Their Applications to Particle Accelerators

Sichen Li, Andreas Adelmann (2022)
arXiv

Automatic setup of 18 MeV electron beamline using machine learning

Francesco Maria Velotti, Brennan Goddard, Verena Kain, Rebecca Ramjiawan, Giovanni Zevi Della Porta, Simon Hirlander (2022)
arXiv

Diagnostics for Linac Optimization With Machine Learning

R. Sharankova, M. Mwaniki, K. Seiya, M. Wesley (2022)
arXiv

Transverse phase space tomography in the CLARA accelerator test facility using image compression and machine learning

Andrzej Wolski, Mark A. Johnson, Matthew King, Boris L. Militsyn, Peter H. Williams (2022)
arXiv

Using Kernel-Based Statistical Distance to Study the Dynamics of Charged Particle Beams in Particle-Based Simulation Codes

Chad E. Mitchell, Robert D. Ryne, Kilean Hwang (2022)
arXiv

Neural Network Solver for Coherent Synchrotron Radiation Wakefield Calculations in Accelerator-based Charged Particle Beams

Auralee Edelen, Christopher Mayes (2022)
arXiv

Adaptive Machine Learning for Time-Varying Systems: Towards 6D Phase Space Diagnostics of Short Intense Charged Particle Beams

Alexander Scheinker, Spencer Gessner (2022)
arXiv

Differentiable Preisach Modeling for Characterization and Optimization of Accelerator Systems with Hysteresis

R. Roussel, A. Edelen, D. Ratner, K. Dubey, J. P. Gonzalez-Aguilera, Y. K. Kim, N. Kuklev (2022)
arXiv

Mixed Diagnostics for Longitudinal Properties of Electron Bunches in a Free-Electron Laser

J. Zhu, N. M. Lockmann, M. K. Czwalińska, H. Schlarb (2022)
arXiv

A Neural Network Model of a Quasi-Periodic Elliptically Polarizing Undulator in Universal Mode

Ryan Sheppard, Cameron Baribeau, Tor Pedersen, Mark Boland, Drew Bertwistle (2022)
arXiv

Physics-informed neural network method for modelling beam-wall interactions

Kazuhiro Fujita (2022)
arXiv

Anomaly Detection in Particle Accelerators using Autoencoders

Jonathan P. Edelen, Nathan M. Cook (2021)
arXiv

Input Beam Matching and Beam Dynamics Design Optimization of the IsoDAR RFQ using Statistical and Machine Learning Techniques

Daniel Koser, Loyd Waites, Daniel Winklehner, Matthias Frey, Andreas Adelman, Janet Conrad (2021)
arXiv

Neural Networks for ID Gap Orbit Distortion Compensation in PETRA III

Bianca Veglia, Ilya Agapov, Joachim Keil (2024)
arXiv

Time-Delayed Koopman Network-Based Model Predictive Control for the FRIB RFQ

Jinyu Wan, Shen Zhao, Wei Chang, Yue Hao (2024)
arXiv

Uncertainty Aware ML-based surrogate models for particle accelerators: A Study at the Fermilab Booster Accelerator Complex

Malachi Schram, Kishansingh Rajput, Karthik Somayaji Peng Li, Jason St. John, Himanshu Sharma (2022)
arXiv

Quantifying Uncertainty for Machine Learning Based Diagnostic

Owen Convery, Lewis Smith, Yarin Gal, Adi Hanuka (2021)
arXiv

Uncertainty Quantification for Virtual Diagnostic of Particle Accelerators

Owen Convery, Lewis Smith, Yarin Gal, Adi Hanuka (2021)
arXiv

Adaptive deep learning for time-varying systems with hidden parameters: Predicting changing input beam distributions of compact particle accelerators

Alexander Scheinker, Frederick Cropp, Sergio Paigua, Daniele Filippetto (2021)
arXiv

Coincident Learning for Beam-based RF Station Fault Identification Using Phase Information at the SLAC Linac Coherent Light Source

Jia Liang, William Colocho, Franz-Josef Decker, Ryan Humble, Ben Morris, Finn H. O'Shea, David

A. Steele, Zhe Zhang, Eric Darve, Daniel Ratner (2025)
arXiv

Using Convolutional Neural Networks to Accelerate 3D Coherent Synchrotron Radiation Computations

Christopher Leon, Petr M. Anisimov, Nikolai Yampolsky, Alexander Scheinker (2025)
arXiv

Explainable physics-based constraints on reinforcement learning for accelerator controls

Colen, Jonathan, Schram, Malachi, Rajput, Kishansingh, Kasparian, Armen (2025)
arXiv

Accelerator system parameter estimation using variational autoencoded latent regression

Mahindra Rautela, Alan Williams, Alexander Scheinker (2024)
arXiv

Long Short-Term Memory Networks for Anomaly Detection in Magnet Power Supplies of Particle Accelerators

Ihar Lobach, Michael Borland (2024)
arXiv

Leveraging Prior Mean Models for Faster Bayesian Optimization of Particle Accelerators

Tobias Boltz, Jose L. Martinez, Connie Xu, Kathryn R. L. Baker, Zihan Zhu, Jenny Morgan, Ryan Roussel, Daniel Ratner, Brahim Mustapha, Auralee L. Edelen (2025)
arXiv

Optimizing Dynamic Aperture Studies with Active Learning

D. Di Croce, M. Giovannozzi, E. Krymova, T. Pieloni, S. Redaelli, M. Seidel, R. Tomás, F. F. Van der Veken (2024)
arXiv

Neural Network Prior Mean for Particle Accelerator Injector Tuning

Connie Xu, Ryan Roussel, Auralee Edelen (2022)
arXiv

A Neural Network approach to reconstructing SuperKEKB beam parameters from beamstrahlung

S. Di Carlo, G. Bonvicini, N. A. Althubiti, R. Ayad, E. De La Cruz-Burelo, I. Domínguez, B. O. El Bashir, H. Farhat, J. Flanagan, R. Gillard, S. Izaguirre Gamez, K. Kanazawa, K. Kumara, D. Liventsev, P. L. M. Podesta-Lerma, D. Ricalde-Herrmann, D. Rodriguez Perez, G. Tejeda-Muñoz, M. Tobiya I. Heredia de la Cruz (2022)
arXiv

Optimizing a Superconducting Radiofrequency Gun Using Deep Reinforcement Learning

David Meier, Luis Vera Ramirez, Jens Völker, Bernhard Sick, Jens Viefhaus, Gregor Hartmann (2022)
arXiv

Uncertainty aware anomaly detection to predict errant beam pulses in the SNS accelerator

Willem Blokland, Pradeep Ramuhalli, Charles Peters, Yigit Yucesan, Alexander Zhukov, Malachi Schram, Kishansingh Rajput, Torri Jeske (2021)
arXiv

Fast, efficient and flexible particle accelerator optimisation using densely connected and invertible neural networks

Renato Bellotti, Romana Boiger, Andreas Adelman (2021)
arXiv

Improving Surrogate Model Accuracy for the LCLS-II Injector Frontend Using Convolutional Neural Networks and Transfer Learning

Lipi Gupta, Auralee Edelen, Nicole Neveu, Aashwin Mishra, Christopher Mayes, Young-Kee Kim (2021)
arXiv

A Novel Approach for Classification and Forecasting of Time Series in Particle Accelerators

Sichen Li, Mélissa Zacharias, Jochem Snuverink, Jaime Coello de Portugal, Fernando Perez-Cruz, Davide Reggiani, Andreas Adelmann (2021)

arXiv

Real-time Artificial Intelligence for Accelerator Control: A Study at the Fermilab Booster

Jason St. John, Christian Herwig, Diana Kafkes, Jovan Mitrevski, William A. Pellico, Gabriel N. Perdue, Andres Quintero-Parra, Brian A. Schupbach, Kiyomi Seiya, Nhan Tran, Malachi Schram, Javier M. Duarte, Yunzhi Huang, Rachael Keller (2021)

arXiv

Surrogate Modeling of the CLIC Final-Focus System using Artificial Neural Networks

J. Ogren, C. Gohil, D. Schulte (2021)

arXiv

Introduction to Machine Learning for Accelerator Physics

Daniel Ratner (2020)

arXiv

Machine learning for design optimization of storage ring nonlinear dynamics

Faya Wang, Minghao Song, Auralee Edelen, Xiaobiao Huang (2019)

arXiv

Studies in Applying Machine Learning to LLRF and Resonance Control in Superconducting RF Cavities

Jorge Alberto Diaz Cruz, Sandra Biedron, Manel Martinez-Ramon, Salvador Sosa, Reza Pirayesh (2019)

arXiv

Machine learning for analysis of plasma driven ion source

N. Joshi, O. Meusel, H. Podlech (2018)

arXiv

First Steps Toward Incorporating Image Based Diagnostics Into Particle Accelerator Control Systems Using Convolutional Neural Networks

A. L. Edelen, S. G. Biedron, S. V. Milton, J. P. Edelen (2016)

arXiv

Neural Networks for Modeling and Control of Particle Accelerators

A. L. Edelen, S. G. Biedron, B. E. Chase, D. Edstrom, S. V. Milton, P. Stabile (2016)

arXiv

Harnessing the power of gradient-based simulations for multi-objective optimization in particle accelerators

Rajput, Kishansingh, Schram, Malachi, Edelen, Auralee, Colen, Jonathan, Kasparian, Armen, Roussel, Ryan, Carpenter, Adam, Zhang, He, Benesch, Jay (2024)

Mach.Learn.Sci.Tech.

BOOSTR: A Dataset for Accelerator Control Systems

Diana Kafkes, Jason St. John (2021)

arXiv

AI-Assisted Transport of Radioactive Ion Beams

Sergio Lopez-Caceres, Daniel Santiago-Gonzalez (2025)

arXiv

Bayesian Optimization Algorithms for Accelerator Physics

Ryan Roussel, Auralee L. Edelen, Tobias Boltz, Dylan Kennedy, Zhe Zhang, Fuhao Ji, Xiaobiao Huang, Daniel Ratner, Andrea Santamaria Garcia, Chenran Xu, Jan Kaiser, Angel Ferran Pousa, Annika Eichler, Jannis O. Lubsen, Natalie M. Isenberg, Yuan Gao, Nikita Kuklev, Jose Martinez, Brahim Mustapha, Verena Kain, Weijian Lin, Simone Maria Liuzzo, Jason St. John, Matthew J. V. Streeter, Remi Lehe, Willie Neiswanger (2024)

arXiv

SPIRAL2 Cryomodules Models: a Gateway to Process Control and Machine Learning

Adrien Vassal, Adnan Ghribi, François Millet, François Bonne, Patrick Bonnay, Pierre-Emmanuel Bernaudin (2021)

arXiv

Predicting Beam Transmission Using 2-Dimensional Phase Space Projections Of Hadron Accelerators

Anthony Tran, Yue Hao, Brahim Mustapha, Jose L. Martinex Marin (2022)

arXiv

Machine Learning for Orders of Magnitude Speedup in Multi-Objective Optimization of Particle Accelerator Systems

Auralee Edelen, Nicole Neveu, Yannick Huber, Mattias Frey, Christopher Mayes, Andreas Adelmann (2020)

arXiv

Superconducting radio-frequency cavity fault classification using machine learning at Jefferson Laboratory

Chris Tennant, Adam Carpenter, Tom Powers, Anna Shabalina Solopova, Lasitha Vidyaratne, Khan Iftekharuddin (2020)

arXiv

Online tuning and light source control using a physics-informed Gaussian process

Adi Hanuka, J. Duris, J. Shtalenkova, D. Kennedy, A. Edelen, D. Ratner, X. Huang (2019)

arXiv

Accurate and confident prediction of electron beam longitudinal properties using spectral virtual diagnostics

Adi Hanuka, C. Emma, T. Maxwell, A. Fisher, B. Jacobson, M. J. Hogan, Z. Huang (2020)

arXiv

Symplectic machine learning model for fast simulation of space-charge effects

Wan, Jinyu, Qiang, Ji, Hao, Yue (2025)

Phys.Rev.Accel.Beams

Application of ensemble machine learning algorithms and filtering techniques in slow orbit feedback systems of electron storage rings

Fan, Jiaqi, Liu, Weibin, Wang, Jiuqing, Wei, Yanru, Wei, Yuanyuan, Ji, Daheng (2025)

Phys.Rev.Accel.Beams

Reconstructing time-of-flight detector values of angular streaking using machine learning

Meier, David, Viefhaus, Jens, Hartmann, Gregor, Helml, Wolfram, Otto, Thorsten, Sick, Bernhard (2025)

Phys.Rev.Accel.Beams

Deep learning framework for fault detection in accelerators

Piekarski, Michal (2024)

JACoW

Machine learning-based non-destructive measurement of bunch length at FRIB

Wan, Jinyu, Plastun, Alexander, Ostroumov, Peter (2024)

JACoW

Beam emittance and Twiss parameters from pepper-pot images using physically informed neural nets

Knight, Ian, Mustapha, Brahim (2024)

JACoW

Advancements in backwards differentiable beam dynamics simulations for accelerator design, model calibration, and machine learning

Roussel, Ryan, Edelen, Auralee, Gonzalez-Aguilera, Juan Pablo, Lehe, Remi, Huebl, Axel, Kaiser, Jan, Santamaria Garcia, Andrea, Xu, Chenran, Eichler, Annika, Charleux, Grégoire (2024)
JACoW

Advanced algorithms for linear accelerator design and operation

Ong, Ysabella, Kassandra, Bellan, Luca, Pisent, Andrea, Comunian, Michele, Fagotti, Enrico, Bortolato, Damiano, Montis, Maurizio, Giacchini, Mauro, Carletto, Osvaldo (2024)
JACoW

Adaptive machine learning with hard physics constraints and generative diffusion for 6D phase space diagnostics

Scheinker, Alexander (2024)
JACoW

Towards Agentic AI on Particle Accelerators

Antonin Sulc, Thorsten Hellert, Raimund Kammering, Hayden Hoschouer, Jason St. John (2025)
Journal

Continuous data-driven control of the GTS-LHC ion source at CERN

Kain, Verena, Rodriguez Mateos, Borja, Bruchon, Niky, Hirlander, Simon, Küchler, Detlef (2024)
JACoW

Machine learning-based extraction of longitudinal beam parameters in the LHC

Iliakis, Konstantinos, Karlsen-Bæck, Birk Emil, Trad, Georges, Timko, Helga, Zampetakis, Michail, Argyropoulos, Theodoros (2024)
JACoW

High fidelity numerical modelling and condition monitoring applied to septum magnets at CERN

Kawa, Krzysztof, Szumlak, Tomasz, Kokkinos, Charilaos, Lackner, Friedrich (2024)
JACoW

Updates to Xopt for online accelerator optimization and control

Roussel, Ryan, Kennedy, Dylan, Boltz, Tobias, Baker, Kathryn, Mayes, Christopher, Edelen, Auralee (2024)
JACoW

Breaking new ground in data-intensive science: first insights from the LIV.INNO center for doctoral training

Welsch, Carsten (2024)
JACoW

The reinforcement learning for autonomous accelerators collaboration

Santamaria Garcia, Andrea, Eichler, Annika, Xu, Chenran, Kaiser, Jan, Scomparin, Luca, Schenk, Michael, Pochaba, Sabrina, Hirlander, Simon (2024)
JACoW

Surrogate Model for Linear Accelerator: A fast Neural Network approximation of ThomX's simulator

Emmanuel Goutierre, Christelle Bruni, Johanne Cohen, Hayg Guler, Michèle Sebag (2023)
IPAC 2023 - 14th International Particle Accelerator Conference

From Compact Plasma Particle Sources to Advanced Accelerators with Modeling at Exascale

Axel Huebl, Remi Lehe, Edoardo Zoni, Olga Shapoval, Ryan T. Sandberg, Marco Garten, Arianna Formenti, Revathi Jambunathan, Prabhat Kumar, Kevin Gott, Andrew Myers, Weiqun Zhang, Ann Almgren, Chad E. Mitchell, Ji Qiang, David Grote, Alexander Sinn, Severin Diederichs, Maxence Thevenet, Luca Fedeli, Thomas Clark, Neil Zaim, Henri Vincenti, Jean-Luc Vay (2023)
20th Advanced Accelerator Concepts Workshop (AAC'22)

RBF neural net based classifier for the AIRIX accelerator fault diagnosis

J. C. Ribes, G. Delaunay, J. Delvaux, E. Merle, M. Mouillet (2000)

arXiv

A Living Review Pipeline for AI/ML Applications in Accelerator Physics

Adnan Ghribi (2025)

arXiv

Reinforcement Learning for Accelerator Beamline Control: a simulation-based approach

Anwar Ibrahim, Alexey Petrenko, Maxim Kaledin, Ehab Suleiman, Fedor Ratnikov, Denis Derkach (2025)

arXiv (Cornell University)

Automated beam tuning of the TRIUMF ISAC facility

Shelbaya, Olivier, Jung, Paul, Kester, Oliver, Hassan, Omar (2025)

JACoW

Beam Energy Forecasting using Machine Learning at the CLEAR accelerator

Gilardi, Antonio, Malyzhenkov, Alexander, Petersson, Alfred, Mostacci Andrea, Pollastro, Andrea, Aksoy, Avni, Filippetto, Daniele, Gamba, Davide, Granados, Eduardo, Tangari, Giacomo, Sjobak, Kyrre, Bonnard, Ladislav, Wroe, Laurence, Carranza-García, Manuel, Franek Ondrej, Korysko, Pierre, Corsini Roberto, Rieker, Vilde, Farabolini, Wilfrid, Mazzoni, Stefano (2025)

JACoW

Advancing accelerator science through data-intensive research and training

Welsch, Carsten (2025)

JACoW

Machine learning approach to MDI optimization for 3 TeV c.o.m. Muon Collider

Castelli, Luca (2025)

JACoW

Advances in machine learning inference of dynamic aperture evaluation for the LHC

Montanari, Carlo Emilio, Di Croce, Davide, Van der Veken, Frederik, Giovannozzi, Massimo, Appleby, Robert, Redaelli, Stefano, Pieloni, Tatiana (2025)

JACoW

Machine learning for calibration drift forecasting in superconducting RF cavities

Sun Yue, Bellandi Andrea, Eichler, Annika, Richter, Bozo, Diomedede, Marco, Schmidt, Christian, Schlarb Holger, Branlard, Julien, Herrmann, Max (2025)

JACoW

Uncertainty-Quantified Machine Model Construction Using Physics-Informed Gaussian Processes and Bayesian Optimization

Isensee, Victoria, Oeftiger, Adrian, Boine-Frankenheim, Oliver (2025)

JACoW

Automated RF Phase Adjustment for Beam Stabilization in the Fermilab Linac

Chichili, R.R., Sulskis, J.A., Sharankova, R., Vamanan, B., Ravi, S. (2025)

Journal

Fast Adaptive Neural Control of Resonant Extraction at Fermilab

Whitbeck, A., Berlioz, J., Danison-Fieldhouse, K., Hazelwood, K., Khan, M., Mitrevski, J., Narayanan, A., John, J.St., Tran, N., Ji, J., Walter, M. (2025)

Journal

First Experimental Demonstration of Machine Learning-Based Tuning on the PSI Injector 2 Cyclotron

M. Haj Tahar, W. Joho, E. Solodko, M. Bocchio, S. Marquie, M. Busch, A. Barchetti, J. Grillenberger, J. Snuerink, M. Schneider (2025)

arXiv (Cornell University)

Xopt and Badger: a machine learning ecosystem for real-time accelerator control and optimization

Roussel, Ryan, Kennedy, Dylan, Edelen, Auralee, Kuklev, Nikita, Miskovich, Sara, Zhang Zhe (2025)
JACOW

Long short-term memory of recurrent neural network to the analysis of temperature rise on the production target in Hadron Experimental Facility of J-PARC

Agari Keizo, Sato Yoshinori, Toyoda Akihisa, Morino Yuhei, Akiyama Hironobu (2025)
JACOW

Autonomous Pressure Control in MuVacAS via Deep Reinforcement Learning and Deep Learning Surrogate Models

Rodriguez-Llorente, Guillermo, Gallardo, Galo, Navascués, Rodrigo Morant, Petrovsky, Nikita Khvatkin, Sabogal, Anderson, Martín, Roberto Gómez-Espinosa (2025)
Journal

Machine learning-based prediction of magnet errors in storage ring light sources

Xu, Jianhao (2025)
arXiv

U-Net-based surrogate modeling for attosecond X-ray free-electron lasers

Yufei Wei, Bingyang Yan, Chenzhi Xu, Jiawei Yan, Haixiao Deng (2026)
arXiv

Towards continual machine learning for particle accelerators

Rajput, Kishansingh, Schram, Malachi, Blokland, Willem, Zhukov, Alexander, Lin, Sen (2026)
JACoW

JuTrack, a Julia-based auto-differentiable accelerator simulation code for advanced dynamics, scientific machine learning and optimization

Wan, Jinyu, Alamprese, Helena, Ratcliff, Christian, Qiang, Ji, Hao, Yue (2026)
JACoW

High-Resolution Image Reconstruction with Unsupervised Learning and Noisy Data Applied to Ion-Beam Dynamics for Particle Accelerators

Francis Osswald, Mohammed Chahbaoui, X. Liang (2026)
Journal of Instrumentation

Machine learning-enhanced infrared imaging for temperature anomaly detection in power supplies

Mohsen, Osama, Lobach, Ihar, Borland, Michael, Sun, Yine (2026)
JACoW

Anomaly detection of slow-moving variables at LANSCE for improved beam quality

Huang, En-Chuan, Yampolsky, Nikolai (2026)
JACoW

Online optimizations of NSLS-II Linac and Linac-to-Booster beam lines using machine learning methods

Song, Minghao, Wang, Guimei, Hidaka, Yoshiteru, Yang, Xi (2026)
JACoW

Machine learning-driven computations of 3D Coherent Synchrotron Radiation

Leon, Christopher, Scheinker, Alexander, Yampolsky, Nikolai, Anisimov, Petr M. (2026)
JACoW

Safe extremum seeking for real-time adaptive accelerator control

Williams, Alan, Scheinker, Alexander (2026)
JACoW

Machine learning assisted Bayesian calibration of accelerator digital twin from orbit response data

Lin, Weijian, Kelly, Christopher, Hoffstaetter, Georg, Brown, Kevin, Urban, Nathan (2026)
JACoW

Integrating simulation and machine learning for Proton Storage Ring beam analysis

Leon, Christopher, Huang, En-Chuan, Scheinker, Alexander, Anisimov, Petr M. (2026)

JACoW

Towards differentiable beam dynamics modeling in BLAST/ImpactX

Huebl, Axel, Mitchell, Chad, Lehe, Remi, Charleux, Grégoire, Myers, Andrew, Zhang, Weiqun, Qiang, Ji, Vay, Jean-Luc, Kaiser, Jan, Hespe, Christian, Gonzalez-Aguilera, Juan Pablo, Xu, Chenran, Santamaria Garcia, Andrea, Roussel, Ryan, Edelen, Auralee, Moses, William Steven (2026)

JACoW

A self-supervised transformer for RF cavity signal denoising

Rajesh, Vikshar, Edelen, Auralee, Hall, Christopher, O'Shea, Finn, Edelen, Jonathan (2026)

JACoW

Machine learning using beam loss monitors for DIAMOND-II

Lehmann, Corey, Bobb, Lorraine, Burrows, Philip (2026)

JACoW

Beam diagnostics, data acquisition system, and applications of machine learning at the KEK e-/e+ Linac

Miyahara, Fusashi, Kinoshita, Kaito, Kurata, Masakazu, Iwasaki, Masako, Natsui, Takuya, Okayasu, Yuichi, Satake, Itsuka, Satoh, Masanori, Uemura, Kosuke, Wang, Di (2026)

JACoW

Advancing accelerator virtual beam diagnostics through latent evolution modeling: an integrated solution to forward, inverse, tuning, and UQ problems

Rautela, Mahindra, Scheinker, Alexander (2026)

arXiv

Machine learning at the Spallation Neutron Source accelerator and target

Armen Kasparian, Willem Blokland, Alexander Zhukov, Anant Ray, Drew Winder, Kishansingh Rajput, Malachi Schram (2026)

JACOW

Machine learning-enhanced deterministic controls in lasers and accelerators

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